Northstar CRM Capstone Project

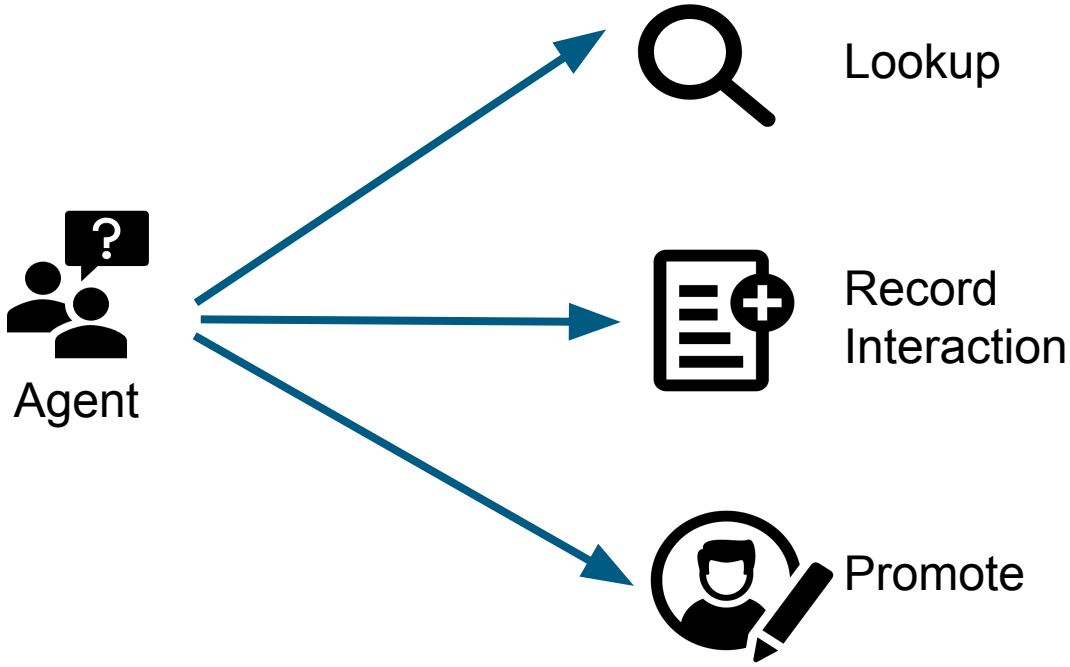
Team C: API-Allstars

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Problem Statement

Banking agents need to:

- **Lookup** customers
- **Record** new Interactions
- **Promote** prospective customers



Live Demo

How we built it

Layer	Choice
Frontend	Angular 22, Node 22 LTS, TypeScript 6.0
API	Spring Boot 4.1.1 on Java 21
Persistence	PostgreSQL 18, Spring Data JPA, Flyway
Messaging	Apache Kafka 4.3
Runtime	k3s
CI/CD	GitHub Actions
Infrastructure	Terraform

What decisions did we make to build this?

ADR 0001 **GitHub Actions** drives CI/CD

ADR 0002 **PostgreSQL** saves data

ADR 0003 **k3s** handles deployment

ADR 0004 **Vertical slices** structures the code

We did away with layer based because…

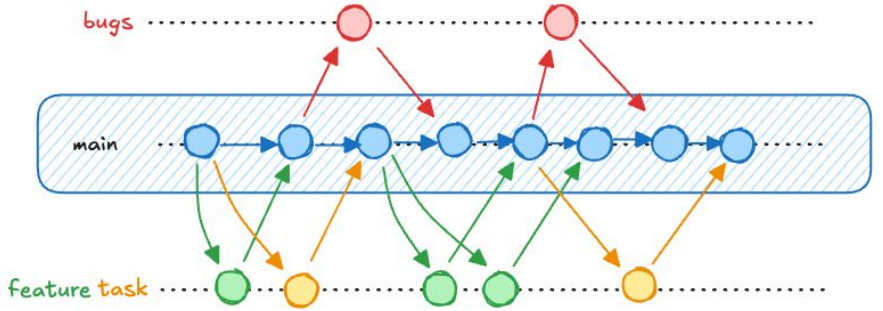
- src/main/java
  - com/northstar/crm
    - controller
    - service
      - ActivateCustomerService.java
      - BrowseCustomersService.java
      - BrowsePublicCustomersService.java
      - RecordCustomerInteractionService.java
      - ShowCustomerTimelineService.java
      - ViewCustomerService.java
      - …
    - repository
    - dto
    - domain
    - platform
    - …

Vertical slices align software with business

- src/main/java
  - com/northstar/crm
    - activateCustomer
      - ActivateCustomerController.java
      - ActivateCustomerService.java
      - ActivateCustomerRepository.java
      - dto
    - viewCustomer
      - ViewCustomerController.java
      - ViewCustomerService.java
      - ViewCustomerRepository.java
      - dto
    - domain
    - platform
    - …

Working agreements improve developer experience

Leveraged trunk-based development with many small pull requests to main



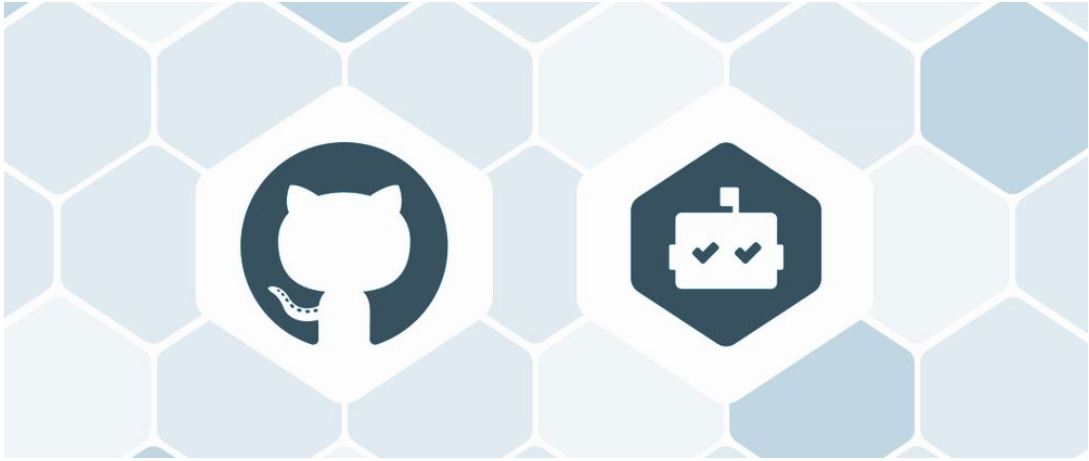
Clean and searchable version history using conventional commits



Source: https://dev.to/glnurltn/trunk-based-vs-branch-based-development-which-strategy-fits-your-team-best-53hn

What challenges did we encounter

- CD dependency checker taking too long
  - PR taking too long
  - Turned off Dependency check scanner in the CI/CD pipeline
  - Used Dependabot



What would we do next

- Email notifications
  - Ex. “Thank you for signing up” to the customer
- Improved observability with dashboards
- Fleshed out authentication system
- Soft delete: the ability to close customer accounts

Q&A